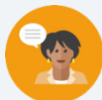


## 9.6 Solving Quadratic Equations by Completing the Square

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                      |                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmarks          | MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.<br>MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.          |                                                                                                      |  Learning Targets                                                      | <ul style="list-style-type: none"><li>I can solve a quadratic by completing the square.</li><li>I can interpret solutions to quadratics in a real-world context as viable or nonviable.</li></ul> |
|  Benchmark Coherence | Building On<br>MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$ , where $p$ is a whole number and $q$ is an integer, determine the real solutions.                                                                                                                                                               |                                                                                                      | Working Toward<br>MA.912.AR.3.2 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system. |                                                                                                                                                                                                   |
|  Essential Question  | <ul style="list-style-type: none"><li>How does the process of completing the square relate to finding the solutions to a quadratic equation using the square root?</li></ul>                                                                                                                                                         |                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                   |
|  Lesson Narrative    | In this lesson, students continue their journey of learning to solve quadratic equations. Students will be introduced to a different method for solving quadratics by completing the square. Students will explore the functionality of completing the square and learn the formal process and steps for using it to find solutions. |                                                                                                      |                                                                                                                                                           |                                                                                                                                                                                                   |
|  Component Summary | 9.6.1 Warm-Up (~5 Min)                                                                                                                                                                                                                                                                                                               | Algebra tiles activity ( <i>SE p. 89</i> )                                                           | 9.6.3 Guided Practice (10 Min)                                                                                                                            | Practice with solving quadratic equations by completing the square ( <i>SE p. 91</i> )                                                                                                            |
|                                                                                                       | 9.6.2 Guided Instruction (15 Min)                                                                                                                                                                                                                                                                                                    | Solving quadratic equations by completing the square ( <i>SE p. 89</i> )                             | 9.6.4 Your Turn! (10 Min)                                                                                                                                 | Independent practice on solving quadratic equations by completing the square ( <i>SE p. 92</i> )                                                                                                  |
|                                                                                                       | 9.6.5 Wrap-Up (~5 Min)                                                                                                                                                                                                                                                                                                               | Student self-reflection activity regarding their perseverance through the lesson ( <i>SE p. 93</i> ) |                                                                                                                                                           |                                                                                                                                                                                                   |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                             |                                                                                                      | Additional Preparation                                                                                                                                    |                                                                                                                                                                                                   |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Completing the square</li></ul>                                                                                                                                                                                                          |                                                                                                      | <ul style="list-style-type: none"><li>(Optional) Prepare small bags with sets of algebra tiles for students to use.</li></ul>                             |                                                                                                                                                                                                   |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>• Students may forget to set the equation equal to 0.</li><li>• Students may forget to apply the addition or subtraction to both sides of the equation.</li><li>• Students may use the wrong sign when completing the square.</li><li>• Students may forget to distribute the <math>a</math>-value to the value added to create a perfect square trinomial.</li></ul> | <b>Things to Consider</b> <ul style="list-style-type: none"><li>• Students may think that this concept is isolated. Connect this solving method to the square root method they learned in the prior lesson. Show how the form the quadratic equation is in can help determine which method might be most appropriate.</li></ul>                                           |
|                                                                                                      | <b>Literacy and Language Routines</b><br><b>True or False</b><br>9.6.3 Guided Practice provides students with a structured, guided practice for solving quadratic equations by completing the square. At key points in the process, the teacher can present a question from the set, make a statement, and have students determine if it's true or false and share out explanations of their thinking and justification.                 | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.3.1: Mathematical Fluency</b><br>The process of completing the square can result in several steps that require substantial calculations. It will be essential that students are able to maintain accuracy when completing each step and maintain flexibility depending on the given situation. |
|                                                                                                      | <br>Differentiate<br>Instruction                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                           |
| Lesson Notes:                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                           |

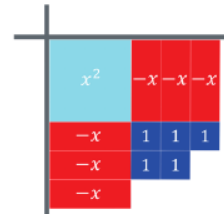
## 9.6.1 Warm-Up: Complete the Square

**Component Overview:** Students will engage in an algebra tile activity that leads them to the conceptual understanding of completing the square. This activity is designed to be done independently. Having quality discussion in this activity will help students connect to the concept.



### 9.6.1 Warm-Up: Complete the Square

The expression  $x^2 - 6x + 5$  is shown.



- How many positive 1-unit tiles should be added to complete the square?  
*4 positive tiles*
- Adding only the positive unit tiles will completely change the original expression, so how many negative 1-unit tiles should be added?  
*4 negative tiles*
- Rewrite  $x^2 - 6x + 5$  in vertex form by placing a number in each square and a sign in each circle.

$$(x \quad \ominus \quad \boxed{3})^2 \ominus \boxed{4}$$



Consider using actual algebra tiles or physical manipulatives to make the same representation to provide kinesthetic entry points. If formal tiles are not available, use paper cut-outs for each of these, or display on a device where applicable.



What key feature of a quadratic graph can be determined by the expression you just wrote?

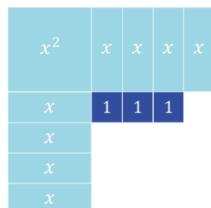
## 9.6.2 Guided Instruction: Solving Quadratic Equations by Completing the Square

**Component Overview:** Students will receive guided instruction on how to solve quadratic equations by completing the square.



### 9.6.2 Guided Instruction: Solving Quadratic Equations by Completing the Square

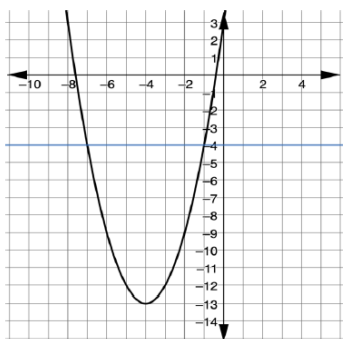
1. The expression  $x^2 + 8x + 3$  is shown.



Rewrite the expression in vertex form.

$$(x + 4)^2 - 13$$

2. The equations  $y = x^2 + 8x + 3$  and  $y = -4$  are shown on the coordinate grid.



Where does the line  $y = -4$  intersect with the quadratic

$$y = x^2 + 8x + 3?$$

$(-7, -4)$  and  $(-1, -4)$

The intersections show the solutions to  $x^2 + 8x + 3 = -4$ .

3. Daniella tried to solve  $x^2 + 8x + 3 = -4$  by using the square root method. Her steps are shown.

$$x^2 + 8x + 3 = -4$$

$$x^2 + 8x = -7$$

$$x^2 = -7 - 8x$$

Daniella got stuck on what to do next. Discuss with your partner why Daniella cannot solve  $x^2 = -7 - 8x$  using square roots.

*Answers will vary. Daniella would have to write the equation in the form  $x^2 = c$ , where  $x^2$  is isolated from constant  $c$ , but has the variable  $x$  on both sides of the equation so that is not possible.*



Use inquiry to drive instruction. Ask questions that elicit connections between the representation of algebra tiles, the graph, and the equation.

- Why is question 1 an expression but question 2 an equation? Can we use algebra tiles for both? Why?
- What do you think the horizontal line represents?
- Which coordinates of the points of intersection are the solution?
- What would the equation need to look like for Daniella to solve using a square root?



Highlight the connection between the way the equation is presented and the method used to solve. Have students share their thinking with the entire class. Students may try to apply the square root method because it is the method they learned just prior to this lesson. Discuss why that does not work in this scenario.

Solving quadratics by completing the square can be shown in different ways. Having students set the quadratic equal to 0 was chosen so that the student follows the same steps for completing the square as they learned in Unit 8, as well as to help students use the quadratic formula when solving quadratics.

## 9.6.2 Guided Instruction: Solving Quadratic Equations by Completing the Square (continued)

4. One way of solving  $x^2 + 8x + 3 = -4$  is by completing the square.

a. Complete the steps in the table.

|                                                     |                                                                         |
|-----------------------------------------------------|-------------------------------------------------------------------------|
| <b>Original Equation</b>                            | $x^2 + 8x + 3 = -4$                                                     |
| <b>Add 4 to both sides to set the equation to 0</b> | $x^2 + 8x + 3 + 4 = -4 + 4$ $x^2 + 8x + 7 = 0$                          |
| <b>Complete the square</b>                          | $(x^2 + 8x) + 7 = 0$ $(x^2 + 8x + 16) + 7 - 16 = 0$ $(x + 4)^2 - 9 = 0$ |

b. Solve  $(x + 4)^2 - 9 = 0$  by completing the steps.

|                                              |                                                 |
|----------------------------------------------|-------------------------------------------------|
| <b>Add 9 to both sides</b>                   | $(x + 4)^2 - 9 = 0$ $(x + 4)^2 - 9 + 9 = 0 + 9$ |
| <b>Take the square root of both sides</b>    | $(x + 4)^2 = 9$ $\sqrt{(x + 4)^2} = \sqrt{9}$   |
| <b>Because <math>\sqrt{a^2} =  a </math></b> | $ x + 4  = 3$                                   |
|                                              | $x + 4 = 3$ or $x + 4 = -3$                     |
|                                              | $x = -1$ or $x = -7$                            |

c. Check your solutions with the intersections found using the graph.

5. Solve  $-3(x - 5)^2 + 6 = -21$ .

$$\begin{aligned}
 -3(x - 5)^2 &= -21 - 6 \\
 -3(x - 5)^2 &= -27 \\
 (x - 5)^2 &= \frac{-27}{-3} \\
 (x - 5)^2 &= 9 \\
 \sqrt{(x - 5)^2} &= \sqrt{9} \\
 |x - 5| &= 3 \\
 x - 5 &= 3 \text{ or } x - 5 = -3 \\
 x &= 8 \text{ or } x = 2
 \end{aligned}$$



Up until this point, how does completing the square to solve compare with completing the square when given an expression?



Challenge students to show the answers are correct in a different way using the graph, a table, or by substituting the values in the equation.

## 9.6.2 Guided Instruction: Solving Quadratic Equations by Completing the Square (continued)

6. Solve by completing the square.

$$-4x^2 - 16x - 11 = -3$$

$$-4x^2 - 16x - 11 + 3 = 0$$

$$-4(x^2 + 4x) - 8 = 0$$

$$-4(x^2 + 4x + 4) - 8 + 16 = 0$$

$$-4(x + 2)^2 + 8 = 0$$

$$-4(x + 2)^2 = -8$$

$$(x + 2)^2 = \frac{-8}{-4}$$

$$(x + 2)^2 = 2$$

$$\sqrt{(x + 2)^2} = \sqrt{2}$$

$$|x + 2| = \sqrt{2}$$

$$x + 2 = \sqrt{2} \text{ or } x + 2 = -\sqrt{2}$$

$$x = -2 + \sqrt{2} \text{ or } x = -2 - \sqrt{2}$$



*Discuss the answer to question 6. Allow students to also find the answer, rounded to the nearest thousandth. Discuss with students why the rounded answer is not the exact answer.*

## 9.6.3 Guided Practice: Solving Quadratic Equations by Completing the Square

**Component Overview:** Students will receive guided practice on how to solve quadratic equations by completing the square.



### 9.6.3 Guided Practice: Solving Quadratic Equations by Completing the Square

1. Solve by completing the square. Show your work.

$$x^2 - 12x + 7 = -17$$

$$\begin{aligned} x^2 - 12x + 24 &= 0 \\ (x^2 - 12x + 36) + 24 - 36 &= 0 \\ (x - 6)^2 - 12 &= 0 \\ (x - 6)^2 &= 12 \\ \sqrt{(x - 6)^2} &= \sqrt{12} \\ |x - 6| &= \sqrt{12} \\ x - 6 &= 2\sqrt{3} \text{ or } x - 6 = -2\sqrt{3} \\ x &= 6 + 2\sqrt{3} \text{ or } x = 6 - 2\sqrt{3} \end{aligned}$$

2. The function  $f(x) = -3x^2 + 54x + 83$  models the sales of music cassette tapes for the years 1980 to 2000, in tens of millions of dollars, where  $x$  is the number of years since 1980.
  - a. A step is shown below in the process of rewriting the function  $f(x) = -3x^2 + 54x + 83$ . Complete the step.  
 $f(x) = -3(x - 9)^2 + 83 + 243$
  - b. The function,  $f$ , models the sales in tens of millions of dollars. How many tens of millions of dollars is \$40,000,000?  
**4 tens of millions of dollars**



Students can leave the square root of 12 as square root of 12. Students can write  $\sqrt{12}$  or  $2\sqrt{3}$ .



Students may need a reference for visualizing music cassette tapes to make meaning of this real-world context.



Discuss with students why they think this number represents the sale of cassette tapes for the years 1980 to 2000. What does  $x = 0$  represent in this context?

## 9.6.3 Guided Practice: Solving Quadratic Equations by Completing the Square (continued)

- c. Solve  $-3x^2 + 54x + 83 = 4$  by completing the square.

$$-3(x - 9)^2 + 326 = 4$$

$$-3(x - 9)^2 = -322$$

$$(x - 9)^2 = \frac{-322}{-3} \text{ or } \frac{322}{3}$$

$$\sqrt{(x - 9)^2} = \sqrt{\frac{322}{3}} \text{ or } \frac{\sqrt{322}}{\sqrt{3}} \approx \frac{17.944358449}{1.73205080757}$$

$$|x - 9| \approx 10.360$$

$$x - 9 \approx 10.360 \text{ or } x - 9 \approx -10.360$$

$$x \approx 9 + 10.360 \text{ or } x \approx 9 - 10.360$$

$$x \approx 19.360 \text{ or } x \approx -1.360$$

- d. Determine which solution is not viable.

$$x \approx -1.360$$

- e. Justify why that solution is not viable.

$x \approx -1.36$  represents a negative number of years, or 1.36 years before 1980. Since the context specifically models years from 1980 to 2000, any solutions before 1980 cannot be viable within the given context.

- f. In what year were the sales of cassette tapes \$40,000,000?

In the year 1999, the sales of cassette tapes were \$40,000,000.



Why is completing the square helpful for this question? Can you think of another method?



Allow students to share their answers to question 2e. Have students agree or disagree with the justifications. Consider allowing students to vote on which solution is non-viable to determine how many students agree and disagree. Have students share their answers and facilitate a dialogue about the rationale for why it's non-viable.



## 9.6.4 Your Turn!

**Component Overview:** Students will work independently to practice solving quadratic equations by completing the square. This activity is designed to be completed independently but can also be done with partner pairs or small groups.



### 9.6.4 Your Turn!

Solve by completing the square. Show your work.

$$\begin{aligned}
 1. \quad & -x^2 + 10x + 8 = -6 \\
 & -(x^2 - 10x) + 14 = 0 \\
 & -(x^2 - 10x + 25) + 14 + 25 = 0 \\
 & -(x - 5)^2 + 39 = 0 \\
 & -(x - 5)^2 = -39 \\
 & (x - 5)^2 = 39 \\
 & \sqrt{(x - 5)^2} = \sqrt{39} \\
 & |x - 5| = \sqrt{39} \\
 & x - 5 = \sqrt{39} \text{ or } x - 5 = -\sqrt{39} \\
 & x = 5 + \sqrt{39} \text{ or } x = 5 - \sqrt{39}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & 6x^2 - 48x - 2 = 4 \\
 & 6x^2 - 48x - 6 = 0 \\
 & 6(x^2 - 8x) - 6 = 0 \\
 & 6(x^2 - 8x + 16) - 6 - 96 = 0 \\
 & 6(x - 4)^2 - 102 = 0 \\
 & 6(x - 4)^2 = 102 \\
 & (x - 4)^2 = 17 \\
 & \sqrt{(x - 4)^2} = \sqrt{17} \\
 & |x - 4| = \sqrt{17} \\
 & x - 4 = \sqrt{17} \text{ or } x - 4 = -\sqrt{17} \\
 & x = 4 + \sqrt{17} \text{ or } x = 4 - \sqrt{17}
 \end{aligned}$$



Consider pulling a small group based on observations or common misconceptions from Guided Practice.



Students may not think about factoring out the  $-1$  in front of the  $x^2$ . Have students who approached question 2 in different ways compare their methods – focusing on their process rather than their product.

## 9.6.5 Wrap-Up: Growth Mindset

**Component Overview:** Students will complete a self-reflection regarding their perseverance through the lesson. This activity should be done independently to allow students to have an honest self-reflection based on their experience and perspective.



### 9.6.5 Wrap-Up: Growth Mindset

"You may encounter many defeats, but you must not be defeated. In fact, it may be necessary to encounter the defeats, so you can know who you are, what you can rise from, how you can still come out of it." — Maya Angelou

1. What was the toughest part of today's lesson for you?  
*Answers will vary. I found the toughest part of today's lesson to be getting the entire calculation correct. I found the toughest part of today's lesson was recognizing when to complete the square.*
2. How did you keep going when things got tough?  
*Answers will vary. I kept working with my partner and asked for help when I needed it. I kept focused on persevering through the problem.*
3. Write a short note to yourself or draw a picture that celebrates your perseverance through the tough part of today's lesson.  
*Answers will vary. When things get challenging, remember that your brain is just working out. It will get stronger with more practice. Stay focused and keep working hard.*



*Universal Design for Learning prioritizes consistent reflection as a practice for meaningful self-regulation. If students are reflecting more often, they are "thinking about their thinking" more often, therefore more likely to engage in this practice independently.*